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REVIEW PAPER

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Mutez Ali Ahmed^{1,2,*}, John Passioura³ and Andrea Carminati¹

¹ Chair of Soil Physics, University of Bayreuth, 95447, Bayreuth, Germany

² Division of Soil Hydrology, Georg-August University of Göttingen, 37077, Göttingen, Germany

³ CSIRO Agriculture and Food, GPO Box 1700, Canberra, ACT 2601, Australia

* Correspondence: mutez.ahmed@uni-bayreuth.de

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Abstract

To a first-order approximation, the yield of a water-limited grain crop depends (i) on how much water is available to the crop and (ii) on how crop water use is partitioned during the growing season. The latter determines the harvest index of the crop, that is, the proportion of the crop's above ground biomass that is converted into grain, which is typically optimal if about 30% of the seasonal available water supply is used during flowering and grain filling. Here, we review the role of roots in extracting water from the soil in both the amount and the timing that may lead to maximal grain yield, and the various mechanisms underlying this activity. These include architectural and anatomical traits; the biophysics of water movement from soil through roots to the leaves including especially the properties of and processes within the interface between roots and soil and the role of mucilage therein; and the physiological role of the roots in influencing the growth and transpiration of the crop canopy, which can optimize the seasonal pattern of water use. These various properties and mechanisms are discussed in the context of improving grain yield in strongly water-limited, especially semiarid, environments.

Keywords: Aquaporins, drought, root anatomy, root architecture, root hairs, root hydraulics, root signals, root–soil interface, mucilage.

Introduction

Inadequate amounts of water can limit the yield of grain crops in many ways. In semiarid environments, such as those with Mediterranean climates, water-limited potential yield (Y_w , the yield of a grain crop when only water is limiting; Fischer *et al.*, 2014) is largely determined (i) by the seasonal amount of available water, and (ii) by how a crop partitions the use of that water during the growing season: a water-limited crop needs an adequate supply of water in the soil at around the time of flowering, roughly about 30% of the total supply, if it is to produce a substantial amount of grain (Passioura and Angus, 2010). Short spells of drought, especially around the

time of flowering, can also prevent Y_w from being achieved. Such spells can markedly affect grain yield not only in semiarid environments, but also in temperate subhumid climates such as those in the most productive cropping areas of Europe where rainfall is typically adequate to produce very high yields.

Much of the laboratory-based literature that deals with the hydraulic properties of roots and soil in relation to 'drought resistance' focuses on the roots being able to take up soil water more rapidly. That may be pertinent in field environments in which the soil profile is rarely depleted of available water. However, in strongly water-limited environments, especially

those that involve terminal droughts, the reverse may be true. Saving water during vegetative growth for use during flowering and grain filling becomes of great importance. This is true of a wide range of strongly water-limited cropping environments in which average grain yields are less than half of those of comparable irrigated crops (Fischer *et al.*, 2014).

Farmers and agronomists, at least in developed countries, are strongly aware of the potential requirement for saving water for later use and can modify their practices to help meet that requirement. A common example is the use of minimal nitrogen fertilizer at the time of sowing to reduce the rate of development of leaf area and hence transpirational water use; more fertilizer can then be supplied if required later in the season (Hunt, 2017).

While astute selection of appropriate genotypes coupled with improved agronomic management has resulted in substantial increases in grain yield in water-limited environments over recent decades (Richards *et al.*, 2014), there is still scope for improving the ability of root systems to capture scarce water especially during flowering and grain filling. Accordingly, our aims in this paper are: (i) to explore the roles of root systems in making best use of a crop's available water supply—essentially the water in the soil at the time of sowing plus the rainfall during the crop's life—in enabling the crop to optimize its development to set an adequate number of seeds per square meter and then to fill those seeds, and (ii) to explore the biophysical processes that are involved in how roots take up water, including the processes in the rhizosphere surrounding those roots, for these processes illuminate how root systems work and could be managed.

The role of the root system in sending growth inhibitory and antitranspirant signals to the leaves is now well-known (Jackson, 2002; Passioura, 2002; Dodd, 2005), at least phenomenologically. The signals can be thought of as feedforward responses to at least part of the soil being too dry, too hard, too cold or too wet. They can modulate the growth of a crop, and hence its water consumption, by reducing the rate of leaf growth and/or stomatal conductance. The mechanisms involved remain controversial, but the knowledge of these effects can result in agronomic improvements, as, for example, in helping to overcome the poor vigor of young crops during the early days of conservation agriculture (Kirkegaard *et al.*, 1995).

These inhibitory signals reduce the demand for water by the leaves, which may be beneficial if it results in saving water during vegetative growth so that there is enough for use during flowering and grain filling. By contrast, the biophysical processes that influence the uptake of water by roots underlie the ability of roots to provide a crop's canopy with as much of a limited water supply as feasible over the full growing season. There are important agronomic questions that provide a context for focusing on these biophysical processes:

(i) Roots that have grown deeply into a moist subsoil often fail to extract all of the seemingly available water there even when the root length density seems to be adequate to take up that water (Passioura, 1991; Kirkegaard *et al.*, 2007; Christopher *et al.*, 2008). Are there hydraulic or osmotic problems that restrict the flow of water from the soil through the rhizosphere into the root?

(ii) Can changes in the hydraulic conductance of the roots and the rhizosphere help improve yield? High conductivity may help plants grow fast by ensuring that leaf water potentials remain high enough to maintain open stomata and to maintain leaf growth. This may be beneficial if there is a large enough supply of water to ensure that the plants do not run out during flowering and grain filling. By contrast, low conductivity may lower the leaf water potential so that the stomata close during periods of large evaporative demand, as in the afternoons of hot days, so that the plants conserve enough water in the soil for use during flowering and grain filling. Given that crops are especially sensitive to water stress during flowering, is it conceivable that the hydraulic conductivity of the root–soil system could be increased at this time thereby raising the water potential of the shoot?

In what follows, and in the context of these questions, we review research on: (i) the role of root architectural traits in determining the rate of water uptake from drying soil; (ii) the role of different root anatomical traits associated with water acquisition from drying soil; and (iii) the complexities of the root soil interface, including the hydraulic influence of mucilage exuded by the root, the possible influence of large concentrations of solutes on the local water potential, the role of root hairs, and possible implications of poor contact with the soil when roots are confined to pre-existing continuous macropores in the subsoil.

Root traits influencing the uptake of soil water in a water-limited environment

Water-limited potential yield, Y_w , the yield of a grain crop when only water is limiting (Fischer *et al.*, 2014), depends on both how much water the crops transpire and the seasonal pattern of that transpiration. The latter is, if anything, more important than the former because it determines the harvest index (HI) of a crop, that is, the ratio of grain yield to above-ground biomass, which can be zero if there is no available water left in the soil at the time of flowering and the crop is maturing during a terminal drought. To a good approximation, yield from a given water supply is maximized if about 30% of the transpiration occurs after flowering (Passioura and Angus, 2010). Figure 1 illustrates the potential importance of conservative behaviour using, for simplicity, the example of rabi crops in south Asia (Fischer *et al.*, 2014). These crops are grown in the dry season between successive monsoon seasons and have to rely largely on water stored in the soil at the time of sowing, water that had accumulated in the soil during the preceding monsoon.

Root architecture

Root architecture is the spatial arrangement and connection of the roots, including the pattern of branching. It plays an important role in determining Y_w . It strongly influences the location of water and nutrient uptake by roots (Passioura, 1983; Ho *et al.*, 2005; Lynch, 2013; Leitner *et al.*, 2014; Lobet *et al.*, 2015; Meunier *et al.*, 2016; Ahmed *et al.*, 2016b). There is

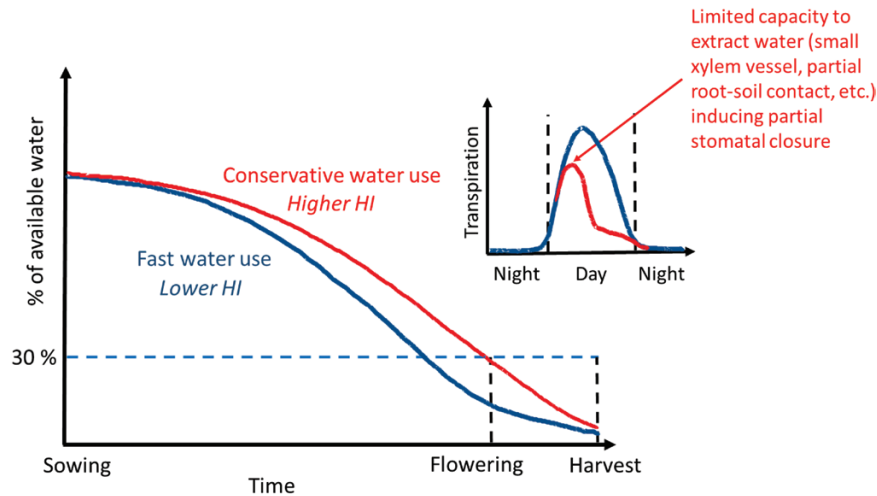


Fig. 1. Time course of water use by a crop that has all of its water supply in the soil at the time of sowing. Conservative water use during the vegetative phase leaves enough water in the soil at the time of flowering to produce a large harvest index (ratio of grain yield to above ground biomass). Fast early water use results in a low harvest index because not enough water is available to produce an adequate number of well-filled grains. The inset illustrates a physiological example of conservative behaviour in which the stomata close during hot afternoons when the ratio of transpiration to CO_2 uptake is especially large and hence inefficient (e.g. Shen *et al.*, 2002; Shekoofa *et al.*, 2016). Examples are given of hydraulic changes that can induce stomatal closure. (This figure is available in colour at JXB online.)

no universally ideal root architecture for grain crops because the optimal distributions, laterally and with depth, depend on the properties of the soil and of the climate (see, for example, Guilpart *et al.*, 2017; Leenaars *et al.*, 2018). Roots that will grow to 2 m will not fulfil their promise of capturing water down to 2 m if the soil profile is only 1 m deep.

Nevertheless, where the soil is deep and there is sufficient rain to wet the deep subsoil, capturing that water can boost yield in a variety of crops experiencing terminal droughts (Cortes and Sinclair, 1986; Ho *et al.*, 2005; Kirkegaard *et al.*, 2007; Gao and Lynch, 2016). In a direct example, Kirkegaard *et al.* (2007) used rainout shelters with controlled drip irrigation to vary the amount of available water in the soil at between 1.35 and 1.85 m depth in the soil to demonstrate that small amounts of deep water can greatly increase wheat yield. They showed that an extra 10 mm of subsoil water captured from that deep layer during grain filling increased the yield by 0.6 tons of grain per hectare (20% more than the control), representing an efficiency of about $60 \text{ kg ha}^{-1} \text{ mm}^{-1}$.

Genotypes with a steeper angle of root growth, and hence increased rooting depth for a given elongation rate, have been shown to yield more under terminal drought in maize (Gao and Lynch, 2016) and common bean (Ho *et al.*, 2005). The ‘steep, cheap, and deep’ maize ideotype, which consists of architectural, anatomical (see next section), and physiological traits, seems better able to capture water from deep soil (Lynch, 2013).

Replenishment and capture of water in the deep subsoil: the agronomic context

While generating deep roots to capture potentially available water in the deep subsoil is attractive, the water has to be there to be captured, and, in wet seasons, it may not even be needed (Lilley and Kirkegaard, 2008). Figure 2 illustrates these points.

Further to the above, given the often large variability of the annual rainfall in strongly water-limited environments, in wet

years the soil profile may be wet beyond the reach of standard length crop roots. Over the years several such wet events can lead to a build-up of ‘available’ water, say 30–50 mm, just outside the reach of the roots. Making use of that water by agronomic or genetic means initially looks attractive, but capturing it could be merely a one-off opportunity, which could also affect the water supply of following crops (Lilley and Kirkegaard, 2016). Thus, the breeding and agronomic management of roots needs careful thought when dealing with a sequence of crops.

Deep water may be replenished in various ways. In humid environments, the usually ample rainfall ensures that the subsoil remains moist, and so provides a reserve on which deep-rooted crops can draw if there is a period of drought during the growing season.

In summer-rainfall monsoonal environments in which temperate crops are grown during the dry season, such as rabi crops in India, the crops essentially face a terminal drought at sowing. Good summer rains can amply replenish water in the subsoil; the challenge thereafter is to meter that water out during the crop’s life, so that there is enough left in the subsoil at the time of flowering to enable the crops to set and fill grain, as illustrated above in Fig. 1.

In semi-arid environments, such as Mediterranean climates, rain falls mostly during the winter–spring growing season, and usually there is enough rain in winter to replenish the subsoil so that the crops have a useful store for roots to rely on during the late spring when the crops are maturing into a hot dry period, essentially a terminal drought in which yield will typically depend on the amount of water available at flowering, as illustrated in Fig. 2B.

In southern Australia, the growing season is in late autumn to late spring, similar to the Mediterranean but there is also significant summer rain. Farmers have now learnt to conserve that summer rain in the subsoil, but early sowing is necessary

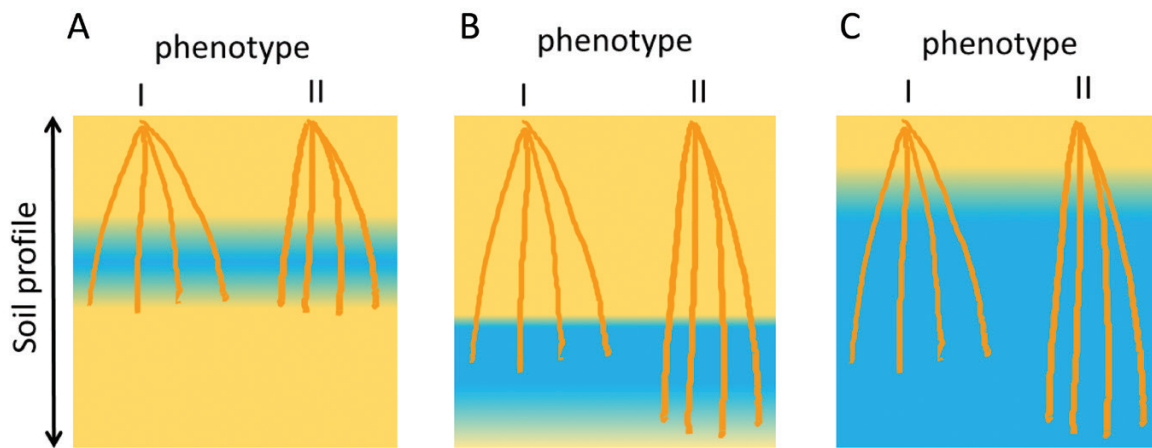


Fig. 2. Root and water distribution at flowering and available during grain filling in (A) a dry season, (B) a moderately good season, and (C) a wet season, for two root types, I and II, of differing potential length. The blue bands signify available water against a backdrop of dry (brown) soil. In (A) the sparse seasonal water supply infiltrates only halfway down the profile and both root types penetrate no further than the wetting front. In (B) the longer root type can access more water than the shorter, and can thereby enable a substantially greater grain yield. In (C) ample water is available to both root types.

to access it most effectively (Hunt, 2017). Early sowing stimulates faster and deeper root growth because of the initially warmer top soil and the longer time available for root growth.

Subsoil water is valuable for several reasons: it is not prone to evaporative losses; knowing how much of it there can improve farmers' risk management; and it is especially valuable during grain filling in crops that experience terminal droughts (Kirkegaard *et al.*, 2007). Yet, the roots of grain crops often fail to extract much of the seemingly available water. The first obvious reason is that roots are not able to grow deep in the subsoil due to compacted soil layers with high penetration resistance. Root ability to penetrate hard soil layers depends on the plant species and is related to root diameter (Clark *et al.*, 2003). However, roots fail to extract subsoil water even when the root length density (length of root per unit volume of soil) seems adequate to do so. For example, with a root length density of only 0.1 cm cm^{-3} , the approximate half-time for the extraction of any remaining available water is only about 3 d if the uptake is predominantly limited by flow through the soil (Passioura, 1991). Even if the roots are restricted to growing in sparse pre-existing continuous macropores, as is commonly the case (Tardieu *et al.*, 1992; White and Kirkegaard, 2010; Hodgkinson *et al.*, 2017), it still seems unlikely that a restricted flow of water through the soil limits the rate of uptake so much that the roots are unable to extract all the available water during a terminal drought over the several weeks of grain filling (Passioura 1991), at least if the flow of water to those roots confined to macropores remains largely radial.

Thus it is likely, in field-grown plants, that the influential hydraulic resistances are at the soil–root interface or within the roots themselves, either radially across the root into the xylem or longitudinally within the xylem, or both (Vadez, 2014).

Morphological and physiological traits of roots that affect the overall hydraulic resistance of a root system

The overall hydraulic resistance of a root system does not in itself influence the transpiration rate of a crop. It can only do so by inducing stomatal closure in the short term (hours) or

by reducing the rate of leaf growth in the longer term (days to weeks), for it is evaporation from the leaves that generates the flow of water through plants. Nevertheless, large hydraulic resistances can have substantial indirect effects as described, especially during hot afternoons when stomatal conductance often falls in response to low leaf water potentials, and in doing so conserves water for later and more productive use.

Longitudinal flow and use of water from the deep subsoil
 Root systems are converging networks. Flows through the various levels of lateral roots converge to generate ever increasing flow rates (volume/time) in the main roots (tap roots in dicots, nodal or seminal axes in monocots), until they meet the shoots of the plants. Because cereal plants (monocots) have no secondary thickening and hence are unable to replicate xylem vessels, the speed of water in the axes accelerates as it moves towards the leaves, and the pressure gradients increase accordingly (Landsberg and Fowkes, 1978), thereby creating a bottleneck.

Temperate cereal crops (such as wheat and barley) have only one, centrally placed, xylem vessel in each of their seminal axes (though several in each nodal axis) and the diameters of those vessels range from about 50 to 60 μm (Richards and Passioura, 1981). Resistance to flow in such small vessels is large. Because this resistance is inversely proportional to the fourth power of the diameter (Poiseuille's law), a small decrease in the diameter, say about 20%, would double the resistance.

Remarkably, wheat plants can grow to maturity with only one seminal root colonizing the soil both in pots (Passioura, 1972) and in the field (Passioura, unpublished). Despite such plants having only one xylem vessel to carry all of the transpiration stream into the shoot, cavitation presumably never occurs for, if it did, the plants would surely die—but they don't. At high transpiration rates, equivalent to about 5 mm d^{-1} in the field, the calculated peak velocity of water in the central xylem vessel as it nears the shoot would be almost 1 m s^{-1} and would require gradients in pressure (calculated from Poiseuille's law) of about 10 kPa mm^{-1} .

Richards and Passioura (1989) conducted a breeding programme to reduce the diameter of the central xylem vessels in

seminal axes and produced selected lines with average diameters of these vessels that were about 15% narrower than unselected lines. In field experiments over 5 years in winter rainfall environments, the selected lines yielded about 5% more, on average, than the unselected ones in dry years, but there were no significant differences between them in wet years. The lack of difference in wet years accorded with expectations, for in such years the development of nodal roots is vigorous and the role of seminal roots in deferring water use becomes irrelevant. Although it has not yet been tested, a high hydraulic resistance in the seminal axes may be of substantial benefit in summer rainfall environments where almost all of the water available to a winter–spring crop would be in the soil at the time of sowing. Deferring a substantial proportion of that water for use during flowering and grain filling is thus essential.

There is considerable interest in breeding for deep roots either empirically (e.g. [Wasson et al., 2012](#)) or physiologically ([Lynch et al., 2014](#); [Chimungu et al., 2014, 2015](#)). The latter has involved selecting genotypes of tropical maize in which the cortices of the roots have reduced metabolic activity because of fewer cells or substantial aerenchyma (large air spaces). These genotypes grow deeper roots seemingly because the saved resources are more accessible to them than to the leaves. Their better performance in this respect has been shown in both controlled environments and in the field.

In temperate crops such as wheat and canola, which grow predominantly during winter and spring in semi-arid environments, deep roots can be engineered agronomically by sowing the crops substantially earlier than is traditional. Doing so enables the roots to grow faster, because the soil is still warm from the summer, and longer, because the roots have more time to grow before the crops flower at the optimal time, which remains the same ([Hunt, 2017](#)); the roots grow 20% deeper and can therefore access any available water in the deep subsoil, thereby potentially increasing the grain yield.

Radial flow from the root surface to the xylem

Water flows radially from the root surface to the xylem vessels crossing several compartments: the cortex, where water can flow in the apoplast and across cells, the endodermis, where the apoplastic flow is impeded and which is the main resistance to the radial flow (see, for example, [Rincon et al., 2003](#)), and the stele. Aerenchyma has been observed in the cortex of many crop roots and has been proposed to decrease the root cost and thus enhance root exploration of the soil resources ([Chimungu et al., 2015](#)). Compared with root architectural traits and with the xylem vessel traits, whose impact on yield have been shown, there are few, if any, examples testing the agronomic relevance of traits, such as aerenchyma, affecting the radial conductivity of roots. Existing studies have been mostly performed in laboratory conditions, mainly because measuring the radial conductivity of roots is not trivial and is often done in systems in which the access to roots is easy, such as hydroponic or sandy substrates. For the moment, possible agronomical implications of traits related to the root radial conductivity remain largely speculative, despite this overall radial conductance being typically much smaller than longitudinal conductance. Nevertheless, the interactions between

radial and longitudinal conductance are of considerable interest in the convergent network of a root system ([Meunier et al., 2018](#)), and may eventually lead to useful agronomic insights, especially in heterogeneous environments.

Also of considerable interest is the potential role of aquaporins in modulating the radial conductivity of roots and how such modulation might influence the uptake of water ([Suku et al., 2014](#); [Chaumont and Tyerman, 2014](#); [Caldeira et al., 2014](#)). No widespread patterns are yet discernible, and other experiments on overall hydraulic conductance of whole plants (e.g. [Munns and Passioura, 1984](#)) have shown little variation in this conductance.

[Caldeira et al. \(2014\)](#) found that root aquaporins (AQP) follow a circadian rhythm, with largest oscillations in plants that had previously been water stressed, and observed that such oscillations were related to plant water status. The authors proposed that AQP affected the root hydraulic conductance, increasing its value at night and early morning, when the transpiration rate is minimum, while decreasing it in the afternoon when the transpiration demand is maximum. The authors formulated the hypothesis that the reduction of root conductance before the peak in transpiration is reached functionally triggers a partial closure of stomata that avoids the rhizosphere becoming water depleted. As soil processes are non-linear, a moderate reduction of water fluxes in the soil would result in a significant increase in plant water potential. Note that this conclusion originated from a model and the impact of such root hydraulic dynamics on yield in water-limited regions remains speculative.

Biophysical processes that affect the flow of water in the soil and in the rhizosphere

As mentioned earlier, roots of annual crops frequently fail to extract all of the seemingly available water in the subsoil by the end of the growing season, yet that water can, if used, substantially increase grain yield. [Figure 3](#) shows an amplified section of [Fig. 1](#), close to the end of the growing season, that illustrates the gap between ‘available’ water (held at water potentials higher than -1.5 MPa) and ‘accessible’ water, which is the water that crops succeed in taking up by harvest time. It is in this region that limitations to water uptake may be most influential in affecting grain yield, yet where the processes involved are generally the most poorly understood.

This section covers several of these processes, namely: physical constraints to water flow through the soil to the rhizosphere (soil adjacent to the roots whose properties are modified by the roots); the role of root hairs in facilitating water movement; and the influence of mucilage, which can alter the water retention and water fluxes across the rhizosphere.

Radial flow of water from bulk soil to roots

The flux of water across the soil–plant–atmosphere continuum is controlled by stomatal conductance and atmospheric demand, but, to avoid dehydration of the plant tissues, the transpiration rate cannot exceed the maximum flow rates

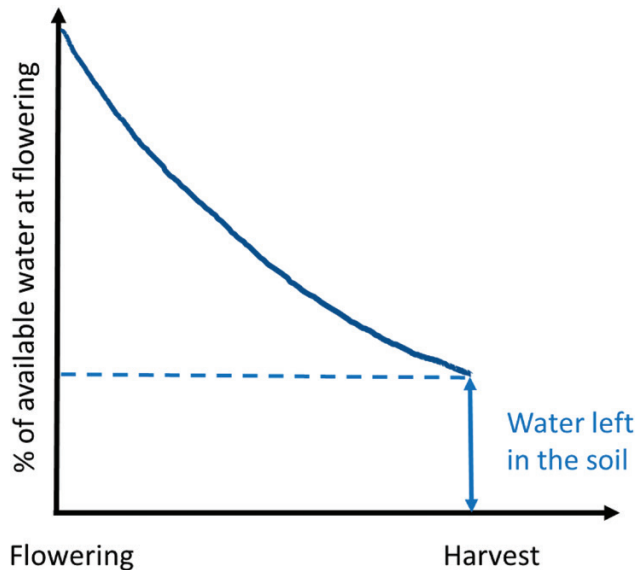


Fig. 3. An amplified section of Fig. 1, close to the end of the growing season. The figure illustrates the gap between ‘available’ water (held at water potentials greater than -1.5 MPa) and ‘accessible’ water, that which crops generally achieve by harvest time. Note that 0% available water corresponds to a soil water potential of *ca* -1.5 MPa. (This figure is available in colour at JXB online.)

that can be sustained by the soil. Water flow to roots is driven by a gradient in matric potential (Ψ_m) between the bulk soil and the root surface. The gradients in matric potential around a single root with cylindrical geometry can be calculated by solving the Richards equation in radial coordinates:

$$\frac{\partial \theta}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r k_{\text{soil}}(\Psi_m) \frac{d\Psi_m}{dr} \right] \quad (1)$$

where θ is the volumetric water content, r is the radial coordinate, t is time, and k_{soil} is the soil conductivity, with the gravitational potential being neglected. Solutions of Eq. (1) show that in wet soils the profiles of water potential and water content around the roots are almost flat, for in wet soil the hydraulic conductivity is typically much larger than the conductivity of the roots (Draye *et al.*, 2010). However, as the soil dries, its hydraulic conductivity drops sharply, by many orders of magnitude. Large gradients in water potential can then occur close to the roots, which in turn reduce the hydraulic conductivity even more. The steep gradients so caused are further steepened because the radial geometry accelerates the velocity of the flowing water as the flow lines converge. These gradients were first calculated by Gardner (1960) and then later shown by Lang and Gardner (1970) and Carminati *et al.* (2011) to result in an absolute limit to the rate of uptake of water by a root. Gardner assumed uniform soil properties around the roots (e.g. that the rhizosphere and the bulk soil had identical hydraulic properties). Thus, in principle, very small hydraulic conductivities of the rhizosphere could strongly limit the uptake of water from drying soil.

However, it is now known that the properties of soil–root interface, and the rhizosphere more generally, may differ greatly from the bulk soil. For example, as the soil dries, roots can shrink, thereby creating air-filled gaps that break

hydraulic continuity between soil particle and the root surface (Huck *et al.*, 1970; Nobel and Cui, 1992). These authors showed that when soil was wet, root conductance was the main influence on hydraulic gradients, but as the soil dries soil conductivity becomes the greater influence. Eventually the formation of air-filled gaps dominates the gradients.

Recently, Carminati *et al.* (2013) investigated the effect of gaps on root water uptake of lupins by simultaneously observing gap formation (using X-ray computed tomography) whilst monitoring the transpiration rate, soil water content, and soil water potential during a drying cycle. They found that gaps formed after the transpiration rate had already fallen and thus concluded that gaps are a consequence and not the initial cause of limited water uptake (Carminati *et al.*, 2013). This makes sense, for if the main resistance to radial flow is at the endodermis (see, for example, Rincon *et al.*, 2003), it follows that the cortex of the root, which is the tissue most likely to shrink, will have a water potential much closer to soil’s than to that of the rest of the plant. Thus it may be somewhat buffered against changes in water potentials experienced by the rest of the plant. Further drying causing the shrinkage of the cortex might also buffer, but for a very short period of time, the inability of the soil to let the water flow to the root. This would temporarily attenuate the decrease in leaf water potential and the effect would be stronger in crops with larger cortex. Such capacitance effects on plant water potential are unlikely to have an impact on the water economy of crops (the water consumed by a crop is much larger than that stored), but they might have some effects on short stress events.

The conclusion that air-filled gaps are not the cause of limited water uptake might not be true for roots growing in the field, which as discussed earlier, are typically found growing in continuous macropores, and with only partial contact with the soil matrix. Thus, the gaps might be a cause of limited water uptake. Indeed, White and Kirkegaard (2010) found that a significant fraction of water in the subsoil (around 20% of available water at maturity) was not extracted by wheat roots, although their root length density seemed to be sufficient for the extraction of water.

An additional process that can limit the water flow at the root–soil interface is accumulation of salts at the root surface, which is likely to occur in drying conditions and at high transpiration rates, when the diffusion of salts is slow and convective fluxes are high. The build-up of solutes would decrease the osmotic potential at the root–soil interface and hence the leaf water potential. Stirzaker and Passioura (1996) argued that the accumulation of solutes at the root–soil interface, which creates the appearance of an additional resistance, might account for the discrepancy in water potential between the leaf and the soil, especially when plants are growing in fertile sandy soils, which are prone to dry rapidly and which may contain substantial concentrations of nutrients in the soil solution (Stirzaker and Passioura, 1996).

Plants have developed ways of coping with steep gradients in water potential at the root–soil interface. Root hairs may help to bridge the gaps between roots and soil and potentially extend the effective root radius for the extraction of water from drying soils, thereby avoiding increasingly

steep gradients in water potential close to the root surface. Mucilage, a polymeric gel exuded at the root tip, maintains the rhizosphere wetter and hydraulically connected to the root surface. In the next sections, we review the role of these two rhizosphere traits on water uptake and speculate on their potential to increase the yield in water-limited regions.

Rhizosphere traits affecting water uptake from drying soil

Root hairs

Root hairs are tubular extensions of epidermal cells of roots. They occur in a defined zone of the root behind the elongation zone and greatly increase the surface area available for the absorption of water and nutrients. Although the role of root hairs in the uptake of immobile elements such as phosphorus (P) is well accepted, their role in water uptake from soil has not been clear.

Dodd and Diatloff (2016) found no difference in the water uptake by the root-hairless *brb* mutant of barley and the wild-type plants grown in drying soil, though they also found that the mutant produced a root system with a much larger fresh weight than that of the wild-type, which might have compensated for its lack of root hairs in taking up water. Segal *et al.* (2008), using magnetic resonance imaging (MRI) on *brb* and wild-type plants, saw a marked depletion of soil water in the root hair zone of the wild-type. However, the differences in the MR images could have been due to uncertainty

of calibration, or to the soil in the rhizosphere having different water-holding properties from the bulk soil as shown by Young (1995).

Carminati *et al.* (2017b) found evidence for the ability of root hairs to take up water from drying soil. They grew barley with and without root hairs (*brb* mutant) in a root pressure chamber (Passioura, 1980), so that the transpiration rate could be varied while the suction in the xylem (a surrogate for leaf water potential) was being monitored. They found that as the soil dried, the xylem suction increased rapidly at high transpiration rates and it did so much more markedly in the hairless mutant, implying a reduced capacity to take up water at high transpiration rates. The hypothesis explaining this difference is illustrated in Fig. 4C.

These experiments also showed strong hysteresis in the relationship between suction and transpiration rate when this rate was reduced, with the recovery of the water potential at the root–soil interface seemingly being remarkably slow. This hysteresis is hard to fathom. It may be related to flow through the rhizosphere, whose properties can differ greatly from those of the bulk soil, or it could be related to transient changes in the hydraulic properties of the roots, possibly because of changes in the activity of aquaporins. Above all, though, it highlights our general lack of understanding of what happens at this interface.

In addition to their role in water and P uptake, Haling *et al.* (2013) demonstrated that root hairs in barley improve root penetration into hard and compacted soils and suggested that root hairs might enable better plant growth under combined

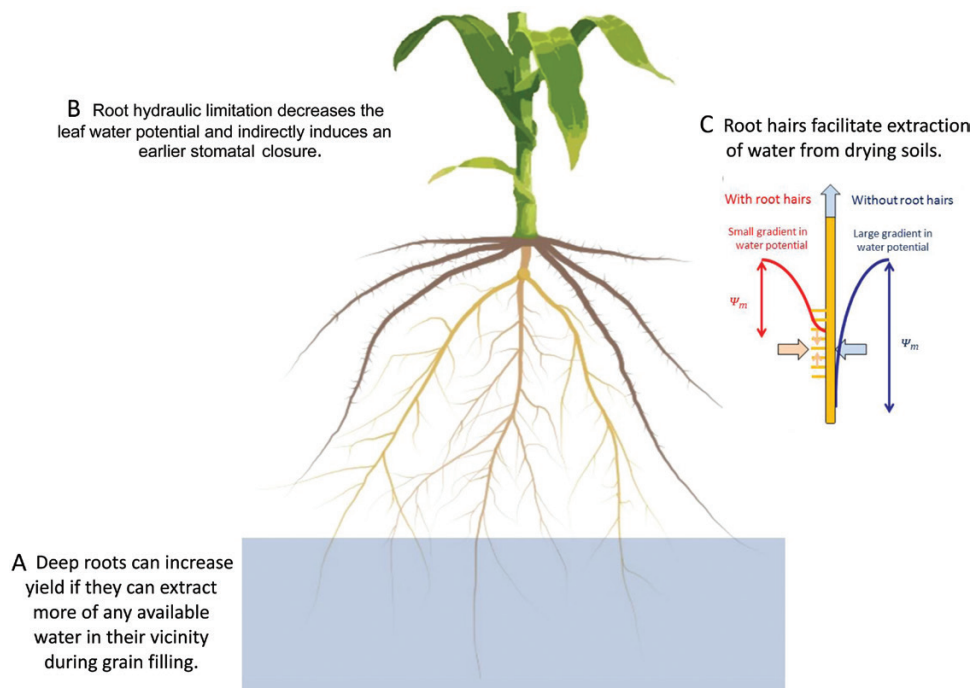


Fig. 4. Summary of the effect of roots and rhizosphere traits on root water uptake and yield from water-limited environments. (A) Deep roots would be beneficial in environments where ample water is available in the subsoil, but only if the roots are able to extract it and if this water is needed in the reproductive stage, when it has a significant impact on the harvest index. (B) During vegetative growth, hydraulic limitation (e.g. due to high resistance in the seminal axes) would be of substantial benefit if it results in soil water being saved for use during grain filling. (C) Root hairs (and possibly mucilage) are expected to decrease the drops in water potential across the rhizosphere and therefore increase the xylem water potential especially at high transpiration rate. Additionally, we expect that root hairs would increase the maximum transpiration rate that can be sustained at a given soil water potential (see Carminati *et al.*, 2017b).

soil stresses. Bengough *et al.* (2016) recently compared a hairless maize mutant with its WT to test the role of root hairs in anchoring the primary root tip during penetration of a sandy loam soil (bulk density 1.0–1.5 g cm⁻³). They found that the wild-type, thanks to the root hairs, had a better ability to penetrate the soil in the range of 1.0 or 1.2 g cm⁻³ soil, but there were no differences between the mutant and the WT in the densest soil (e.g. 1.5 g cm⁻³) (Bengough *et al.*, 2016).

Despite mounting evidence that root hairs favour root growth and extraction of water and nutrients from the soil, and that several genes that control root hair formation and elongation have been identified (Hochholdinger *et al.*, 2008) and pronounced genetic variation in root hair length have been shown in barley (Gahoonia and Nielsen 2004) and maize (Hochholdinger *et al.*, 2008), the impact of root hairs on yield in the field remains unknown for water-limited plants, though well described for uptake of phosphorus (Gahoonia and Nielsen, 2004).

Mucilage

Mucilage is a polymeric gel secreted by the root tip and capable of absorbing large volumes of water, though, in isolated form, only at suctions smaller than about 10 kPa (McCully and Boyer, 1997). These authors also noted that mucilage might bind soil particles together thereby forming a cohesive rhizosheath around the roots. Young (1995) found that the rhizosheath of wheat was wetter than the bulk soil and attributed it to the role of mucilage, which, in interacting with soil particles, could alter the pore size distribution in favour of greater water retention (see also Watt *et al.*, 1994; Carminati *et al.*, 2017a; Koebernick *et al.*, 2017; Helliwell *et al.*, 2017). The low surface tension of mucilage is likely to decrease the soil water content (Read *et al.*, 2003), but this is mainly relevant in the wet range of water potentials. Anyway, the volumes of water retained by mucilage in the rhizosphere are small and would be taken up by plants in a few hours, which would confirm the conclusion of McCully and Boyer (1997) that mucilage *per se* does not provide additional water that could be taken up by crops when the bulk soil is depleted. Probably, the role of mucilage in soil–plant water relations is related to its effects on the hydraulic conductivity of the root–soil interface. Carminati *et al.* (2010, 2011) hypothesized that the enhanced water retention buffers the decrease in hydraulic conductivity of the rhizosphere during soil drying. The authors ran simulations of water uptake by a single root and found that mucilage would facilitate the extraction of water from drying soils by attenuating the drop in water potential across the rhizosphere (with a similar result to that achieved by root hairs). This numerical exercise still needs to be supported by experimental evidence.

An additional property of mucilage affecting water fluxes across the root–soil interface is that it repels water when dry. This has been shown for mucilage exuded by nodal roots of maize (Ahmed *et al.*, 2015, 2016a). Zickenrott *et al.* (2016) showed that contact angle increases with mucilage content, but with high variability among plant species. Rhizosphere water-repellency and its effect on root water uptake is clearly evident in lupins in sandy soils (Moradi *et al.*, 2012; Zarebanadkouki *et al.*, 2016). It was shown to temporarily

limit root rehydration and water uptake upon a cycle of severe drying and rewetting (Zarebanadkouki *et al.*, 2016). It is likely that such an effect is relevant mainly in sandy soils, which have a small specific surface area. It is negligible in fine textured soils, as explained in Benard *et al.* (2018), but it is unclear to what extent this can be generalized across plant species. For instance, Zickenrott *et al.* (2016) found highly variable contact angles of mucilage of varying species. Similarly, Naveed *et al.* (2017) showed that mucilage of maize and barley have contrasting properties, with maize mucilage being more viscous and water repellent. This repellency may limit the ability of roots in dry top soil to recover when rains moisten the soil, so that the roots can extract that water before it evaporates from the soil surface. On the other hand, mucilage repellency might isolate the roots from drying soils and reduce water losses from the roots to the soil. During night-time, hydraulic redistribution would drive water from roots sitting in wet soil regions to roots sitting in dry soil regions. In this way mucilage would be rehydrated, which would avoid this water draining into the bulk soil.

Lynch *et al.* (2014) argued that because mucilage is mainly present in the most distal and young segments and because these regions have been believed not to take up water (Wang *et al.*, 1991), mucilage (and the same holds true for rhizosheath) should have little influence on root water uptake. This conclusion may be true, but the claim that distal roots do not take up water is probably not valid for all plant species and root types, as shown for mature maize, whose nodal roots, in contrast to the seminal roots, take up water also from the distal segments (see Ahmed *et al.*, 2018).

There is some evidence that mucilage facilitates water extraction from drying soils, as shown for an artificial set-up with a suction cup covered with chia seed mucilage and then placed in a dry soil (Ahmed *et al.*, 2014). The facilitated water extraction from drying soils might arise from higher hydraulic conductivity in dry conditions, as a result of the enhanced water retention and continuity of the liquid phase. Carminati *et al.* (2017a) showed that the continuity of the liquid phase in porous media increased with viscosity of the liquid, as the viscosity avoids the break-up of liquid bridges due to capillary forces. However, the mucilage network might eventually decrease the soil permeability, not only in saturated conditions, as shown in Kroener *et al.* (2018), but also in unsaturated conductivity. Alternatively or additionally, the putative enhanced water extraction might arise from the mucilage binding soil particles to the root surface, thereby maintaining the roots mechanically and hydraulically connected to adjacent soil particles. The potential role of mucilage on root water uptake remains hard to discern, but its intriguing properties are hard to dismiss.

In summary, the second section, ‘Root traits influencing the uptake of soil water in water-limited environment’, dealt with root traits that have been useful in improving yield in water-limited environments. However, there remains a major gap in understanding the processes that influence, perhaps sometimes inhibit, water flow across the root–soil interface. The role of root hairs is still uncertain though they may well facilitate the uptake of water by buffering the water content at the surface of the root, as shown in Fig. 4C, thereby maintaining

a higher leaf water potential at high transpiration rates. There is a large and puzzling hysteresis in the relationship between leaf water potential and transpiration rate when this rate is raised then lowered. It is not known whether this is outside the roots or within them.

The effects of mucilage on water uptake are seemingly diverse. It may help, it may hinder, or it may have no effect. It does, however, contribute substantially to the properties of the rhizosphere, and it may eventually turn out to have functions other than as a lubricant for the growth of roots through the soil.

Concluding comments

We have aimed in this paper to discuss the biophysical processes involved in the uptake of water by crop plants, and to do so within the context of a simple but nevertheless informative agronomic framework that focuses on the water economy of a crop. This framework embodies two commonly used agronomic propositions. These are, first, that the aboveground biomass of water-limited crops depends on the amount of water transpired by those crops; and second, that, to maximize grain yield from a given water supply, about 30% of that water should be available to use after about the time of flowering, at least in crops that typically experience terminal droughts. The rationale for the second proposition is that it roughly optimizes the proportion of the biomass that ends up as grain, a ratio known as the harvest index.

The second section of this paper dealt mostly with macroscopic properties of root systems, their architecture and their hydraulic properties, and how these interact to extract water from soil. Most of the literature in this area concentrates on root traits that enable faster uptake of water, though a more pertinent goal is to modulate the rate of uptake during the growing season to ensure that enough of the water supply is available from the time of flowering to enable a crop to produce an adequate number of seeds per square metre and the resources to fill those seeds.

The third section dealt with the biophysics of how water moves from the soil to the vascular tissue of the roots. While the movement of water through the bulk soil is reasonably well understood, its passage through the rhizosphere, across the root–soil interface, and radially across the cortex and endodermis of a root growing in soil, is not. In the context of the water economy of a crop, there seem to be impediments to the uptake of water from the subsoil, water that could contribute greatly to yield if used during the grain-filling period. Our knowledge of the roles in water uptake of root hairs, air gaps, mucilage, and aquaporins in roots growing in soil, is, as yet, rudimentary. The challenge ahead is to explore these processes in field soils. New insights may enable the closing of the gap illustrated in Fig. 3 between the water that crops generally find accessible in the subsoil and the commonly accepted limit of water held at above a water potential of -1.5 MPa. Any extra water uptake during grain filling adds substantially to grain yield.

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